

Central-inversion-invariant domination on the $7 \times 7 \times 7$ queen graph

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Abstract

We determine the minimum size of a dominating set invariant under central inversion in the nonperiodic three-dimensional queen graph on $\{0, \dots, 6\}^3$: it is exactly twelve. The lower bound reduces all invariant sets of size at most eleven to the center and at most five opposite pairs. A Boolean encoding with 1,182 variables and 4,000 clauses is excluded by a complete, checked DRAT refutation. An explicit invariant twelve-queen placement in the middle plane attains the bound. The original problem of determining the unrestricted domination number remains unresolved by this work; the result excludes only the specified invariant class from potential placements below twelve.

1 Statement and context

Let $V = \{0, \dots, 6\}^3$. Two distinct vertices of Q_7^3 are adjacent when their difference is a nonzero integer multiple of a vector in $\{-1, 0, 1\}^3 \setminus \{(0, 0, 0)\}$. Opposite direction vectors define the same line, giving thirteen undirected queen directions. There is no periodic identification. Adjacency depends only on the two cells, without occupancy-dependent blocking. Write $N[q]$ for the closed neighborhood of q ; thus a queen covers its own cell. A set $D \subseteq V$ dominates if $\bigcup_{q \in D} N[q] = V$.

Define

$$I(a, b, d) = (6 - a, 6 - b, 6 - d), \quad \gamma_I(Q_7^3) = \min\{|D| : D \text{ dominates } Q_7^3, I(D) = D\}.$$

Theorem 1. *The central-inversion-invariant domination minimum is $\gamma_I(Q_7^3) = 12$. Consequently, every dominating set of Q_7^3 of size at most eleven fails to be invariant under I .*

Ramani [1] reports the unrestricted interval $10 \leq \gamma(Q_7^3) \leq 12$ in Section 4, Table 1, and provides a twelve-queen placement. The reported lower endpoint comes from a Gurobi computation halted after 48 hours; we neither certify nor use that endpoint. The associated implementation is available as [2]. Theorem 1 instead determines the minimum subject to the additional condition $I(D) = D$. Its lower bound excludes both ten queens arranged in opposite pairs and eleven queens consisting of the center and five pairs, possibilities not excluded by the reported interval.

Requiring a set to be invariant is different from replacing a set by an equivalent placement under a cube automorphism. In particular, our argument does not assume that an arbitrary dominating set can be made invariant. No unrestricted upper-bound improvement, classification of all minimum sets, or result for other board orders is asserted. As adjacent computational methodology, Zeng [3] studies compressed Boolean encodings and checked refutations for a geometric extremal problem. That reference is methodological context, not a source of the queen-graph equality proved here.

2 The lower bound

Reduction to opposite pairs

Central inversion negates coordinate differences, so it is a graph automorphism. Its only fixed vertex is $c = (3, 3, 3)$; the other vertices form 171 two-element orbits. An invariant set therefore has size $2r + \varepsilon$, where r counts selected opposite pairs and $\varepsilon \in \{0, 1\}$ records the center. If its size is at most eleven, then $r \leq 5$. If the center is absent, adjoining it preserves domination and still leaves at most eleven vertices. Conversely, the center and at most five pairs form an invariant set of size at most eleven. Hence

$$\gamma_I(Q_7^3) \leq 11 \iff \text{the center and at most five opposite pairs dominate } Q_7^3. \quad (1)$$

This equivalence covers every invariant set below twelve, including all smaller cardinalities.

Order the vertices by $\iota(a, b, d) = 49a + 7b + d$. For $1 \leq i \leq 171$, let P_i contain the vertices of indices $i - 1$ and $343 - i$, and let the Boolean variable x_i select the entire pair P_i . The center covers exactly $1 + 13 \cdot 6 = 79$ cells. For each of the other 264 cells p , impose

$$C_p = \bigvee_{\{i: p \in \bigcup_{q \in P_i} N[q]\}} x_i. \quad (2)$$

Since each pair is invariant, $C_p = C_{I(p)}$. Retaining the cells p with $\iota(p) < \iota(I(p))$ therefore leaves exactly 132 coverage clauses without losing a condition. All 171 possible pairs are available; this lower-bound encoding imposes no planar restriction.

An exact cardinality counter

For $1 \leq i \leq 171$ and $1 \leq j \leq \min(i, 6)$, introduce $s_{i,j}$ with the intended meaning that at least j of $x_1, \dots, x_i$ are true. Set the boundary constants $s_{i,0} = \text{true}$ and $s_{i,j} = \text{false}$ when $j > i$. Use the recurrence

$$s_{i,j} \iff s_{i-1,j} \vee (s_{i-1,j-1} \wedge x_i). \quad (3)$$

For $a = s_{i-1,j}$, $b = s_{i-1,j-1}$, $x = x_i$, and $y = s_{i,j}$, its conjunctive normal form is

$$(\neg a \vee y) \wedge (\neg b \vee \neg x \vee y) \wedge (\neg y \vee a \vee b) \wedge (\neg y \vee a \vee x). \quad (4)$$

Simplify constant literals, discard tautological clauses, and remove repeated literals within each clause. Finally impose the unit clause $\neg s_{171,6}$.

Induction on i in (3) proves the intended meaning of every counter variable. Every selection of at most five pairs has a counter extension satisfying these clauses, and every satisfying assignment selects at most five pairs. Let F be the conjunction of the 132 clauses (2), the counter clauses, and the terminal unit clause. There are $1 + 2 + \dots + 5 + 166 \cdot 6 = 1,011$ counter variables, so F has 1,182 variables in total and, after the indicated simplifications, 4,000 clauses. We have proved

$$F \text{ is satisfiable} \iff \gamma_I(Q_7^3) \leq 11. \quad (5)$$

The finite refutation

Kissat produced a complete binary DRAT (deletion resolution asymmetric tautology) refutation of F . The saved certificate has 118,258,896 bytes. It was checked with `drat-trim`, and the same complete certificate was replayed in a separate review pass; the checker exited successfully and

reported `s VERIFIED`. This checks a derivation of the empty clause for the precise saved formula. The solver’s UNSAT label alone is not the justification.

The separate verifier reconstructs adjacency by integer coordinate differences, compares it with ray stepping on all $343^2 = 117,649$ ordered pairs, and checks all 264 original coverage rows against the 132 retained rows. It accounts for every CNF clause and checks each of the 1,011 counter gates on every assignment of its distinct local variables (14,764 assignments in total). The general cardinality argument is the induction above, rather than a sample of counter inputs. Thus the checked refutation applies to the graph and the exact class in (1). By (5), $\gamma_I(Q_7^3) \geq 12$.

3 A matching placement and verification scope

For the upper bound, take $D = \{(3, b, d) : (b, d) \in S\}$, where

$$S = \{(0, 3), (1, 0), (1, 6), (2, 2), (2, 4), (3, 1), \\ (3, 5), (4, 2), (4, 4), (5, 0), (5, 6), (6, 3)\}.$$

These are twelve distinct board cells. The displayed list is closed under $(b, d) \mapsto (6 - b, 6 - d)$, so $I(D) = D$. For cells p, q , the exact closed-neighborhood test is

$$p \in N[q] \iff |\{p_k - q_k : 1 \leq k \leq 3\} \setminus \{0\}| \leq 1. \quad (6)$$

Indeed, for distinct cells all nonzero absolute differences must be equal; their signs specify a queen direction. Equal cells satisfy the closed neighborhood condition as well. The following complete integer check verifies the placement without a solver:

```
from itertools import product
S = [(0,3),(1,0),(1,6),(2,2),(2,4),(3,1),
      (3,5),(4,2),(4,4),(5,0),(5,6),(6,3)]
D = {(3,b,d) for b,d in S}
V = set(product(range(7), repeat=3))
assert len(D) == 12 and D <= V
assert {tuple(6-t for t in q) for q in D} == D
def covers(p, q):
    return len({abs(a-b) for a,b in zip(p,q)} - {0}) <= 1
assert all(any(covers(p,q) for q in D) for p in V)
```

All 343 cells pass, also with the separate ray-based implementation; the review verifier records a covering queen for each cell. Hence $\gamma_I(Q_7^3) \leq 12$, completing the proof of Theorem 1. The SAT run that found the placement is discovery evidence; its trace is not used as an unsatisfiability certificate.

Reproducibility and computational assistance. The accompanying computational dossier contains the exact CNF, complete binary DRAT refutation, orbit data, explicit witness, encoder, separate verification programs, and checker records. Its manifest identifies these files and its README gives replay commands. The encoder, exact Python computations, and `drat-trim` checker form part of the trusted computational base. This is a finite computer-assisted proof, with no proof-assistant formalization claimed. AI assistance was used for proof search, code development, a separate mathematical review pass, and manuscript preparation. That review pass is not external peer review or a claim of statistically independent evidence; the mathematical support is the reduction, checked refutation, and exhaustive witness verification.

Literature and remaining scope. The comparison is limited to the preserved version of Ramani’s paper [1] and the inspected associated code [2]. No equality for the invariant minimum was located there. Attempts to refresh the arXiv record and perform a targeted search failed because of DNS access errors. These access limitations do not establish priority or the present worldwide status of the unrestricted problem. The present result supplies an exact benchmark for the invariant class; it leaves the existence of unrestricted ten- or eleven-queen placements undecided.

References

- [1] Mahesh Ramani, *On the Structure of 3D Queen Domination*, arXiv:2604.03793v1, 2026. <https://arxiv.org/abs/2604.03793v1>.
- [2] Mahesh Ramani, *Calculation of optimal solutions to the three dimensional queen’s domination problem*, software, Zenodo, 2026. <https://doi.org/10.5281/zenodo.19412672>.
- [3] Zijian Zeng, *One-Sided Parity Compression for Certified SAT Attacks on the 33-Point Erdős–Szekeres Problem*, 2026. <https://doi.org/10.2139/ssrn.7334758>.